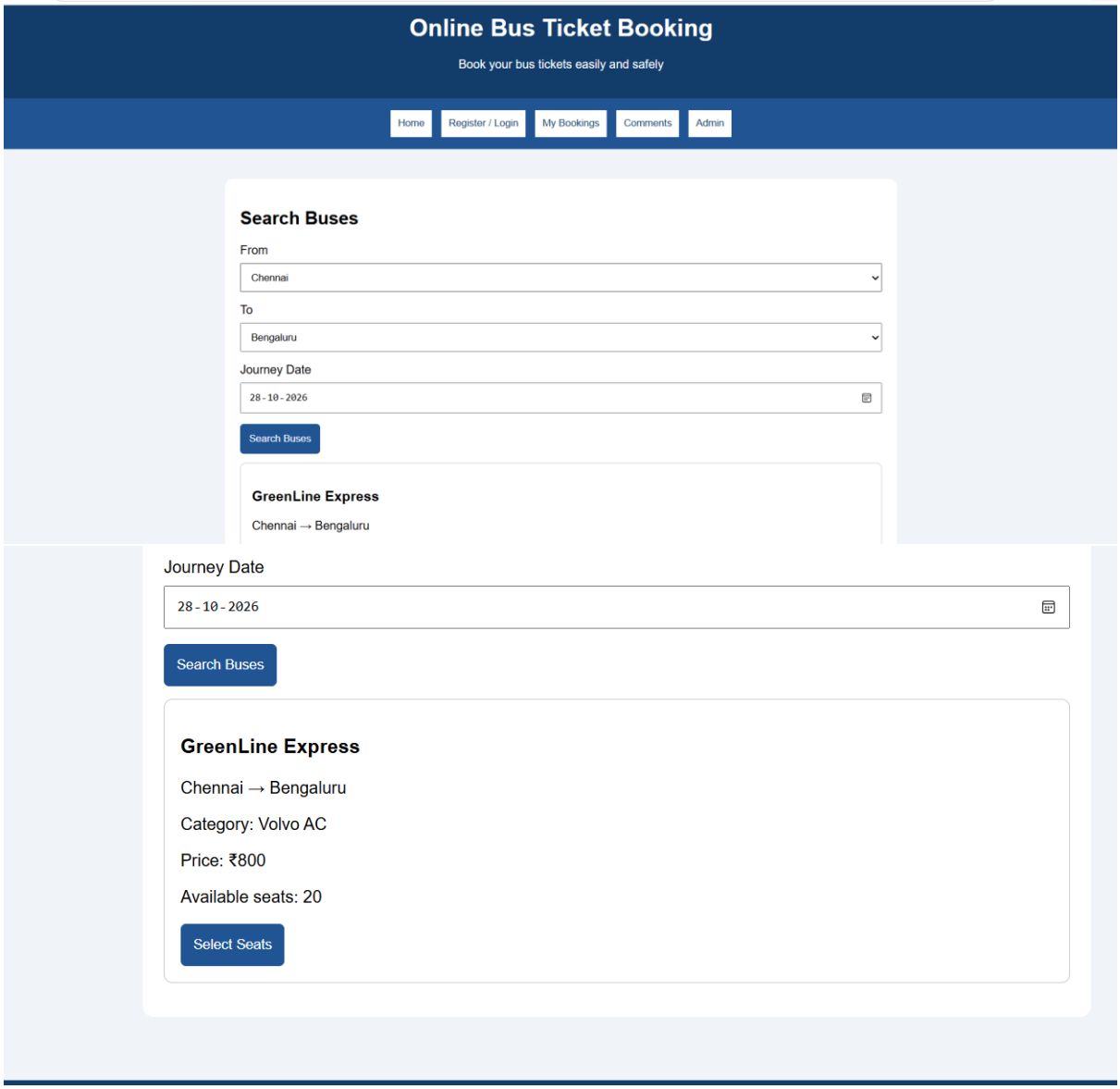
3. The online bus ticket booking system enables users to search for buses by route, date, and time, and view available seats. Users can create an account, log in, and manage their bookings through a personal dashboard. The booking process includes selecting seats, filling out passenger details, and making secure payments via an integrated payment gateway. Admins can manage bus schedules, seat availability, and user accounts through an admin panel. Users receive booking confirmation emails and can view their booking history. The website is mobile-responsive, ensuring easy access on all devices.




```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Bus Ticket Booking</title>

<style>
  body {
    font-family: Arial, sans-serif;
    background: #f0f4f8;
    margin: 0;
  }

  header {
    background: #123c69;
    color: white;
    text-align: center;
    padding: 20px;
  }

  nav {
    background: #205493;
    padding: 12px;
    text-align: center;
  }

  nav button {
    background: white;
    color: #123c69;
    border: none;
    padding: 10px;
    margin: 4px;
    cursor: pointer;
  }
```

```
.container {
   max-width: 900px;
   margin: 25px auto;
   padding: 15px;
}

.box {
   background: white;
   padding: 20px;
   margin-bottom: 20px;
   border-radius: 10px;
}

input, select {
   width: 100%;
   padding: 10px;
   margin: 8px 0 15px;
   box-sizing: border-box;
}

.main-btn {
   background: #205493;
   color: white;
   border: none;
   padding: 12px;
   cursor: pointer;
   border-radius: 5px;
}

.bus {
```

```
    border: 1px solid #ccc;
    padding: 15px;
    margin: 12px 0;
    border-radius: 8px;
}

.seat {
    width: 45px;
    padding: 10px 0;
    margin: 5px;
    background: #b7e4c7;
    border: none;
    cursor: pointer;
}

.seat.selected {
    background: #205493;
    color: white;
}

.seat.booked {
    background: #aaa;
    cursor: not-allowed;
}

.hidden {
    display: none;
}

footer {
    text-align: center;
```

```
        padding: 20px;
        background: #123c69;
        color: white;
      }
    </style>
</head>

<body>

<header>
  <h1>Online Bus Ticket Booking</h1>
  <p>Book your bus tickets easily and safely</p>
</header>

<nav>
  <button onclick="showPage('home')">Home</button>
  <button onclick="showPage('register')">Register / Login</button>
  <button onclick="showPage('bookings')">My Bookings</button>
  <button onclick="showPage('comments')">Comments</button>
  <button onclick="showPage('admin')">Admin</button>
</nav>

<div class="container">

  <!-- HOME AND BUS SEARCH -->
  <section id="home" class="box">
    <h2>Search Buses</h2>

    <label>From</label>
    <select id="from">
      <option>Chennai</option>
```

```
      <option>Bengaluru</option>
      <option>Coimbatore</option>
      <option>Madurai</option>
    </select>

    <label>To</label>
    <select id="to">
      <option>Bengaluru</option>
      <option>Chennai</option>
      <option>Madurai</option>
      <option>Coimbatore</option>
    </select>

    <label>Journey Date</label>
    <input type="date" id="date">

    <button class="main-btn" onclick="searchBuses()">
      Search Buses
    </button>

    <div id="busList"></div>
  </section>

  <!-- REGISTRATION AND LOGIN -->
  <section id="register" class="box hidden">
    <h2>User Registration</h2>

    <input id="name" placeholder="Enter your name">
    <input id="email" type="email" placeholder="Enter email">
    <input id="password" type="password" placeholder="Enter password">
```

```
  <button class="main-btn" onclick="registerUser()">
    Register
  </button>

  <hr>

  <h2>User Login</h2>
  <input id="loginEmail" type="email" placeholder="Email">
  <input id="loginPassword" type="password" placeholder="Password">

  <button class="main-btn" onclick="loginUser()">
    Login
  </button>

  <p id="message"></p>
</section>

<!-- BOOKING HISTORY -->
<section id="bookings" class="box hidden">
  <h2>My Bookings</h2>
  <div id="bookingList"></div>
</section>

<!-- COMMENTS -->
<section id="comments" class="box hidden">
  <h2>Comments and Reviews</h2>

  <input id="commentName" placeholder="Your name">
  <input id="commentText" placeholder="Write your comment">

  <button class="main-btn" onclick="addComment()">
```

```
      Post Comment
    </button>

    <div id="commentList"></div>
  </section>

  <!-- ADMIN PANEL -->
  <section id="admin" class="box hidden">
    <h2>Admin Login</h2>

    <input id="adminUser" placeholder="Username">
    <input id="adminPassword" type="password" placeholder="Password">

    <button class="main-btn" onclick="adminLogin()">
      Login as Admin
    </button>

    <div id="adminPanel" class="hidden">
      <h3>Add Bus Information</h3>

      <input id="busName" placeholder="Bus name">
      <input id="busFrom" placeholder="From">
      <input id="busTo" placeholder="To">

      <label>Bus Category</label>
      <select id="category">
        <option>Volvo AC</option>
        <option>Sleeper Coach</option>
        <option>AC Coach</option>
      </select>
```

```
        <input id="price" type="number" placeholder="Ticket price">
        <input id="totalSeats" type="number" placeholder="Total seats">

        <button class="main-btn" onclick="addBus()">
          Add Bus
        </button>

        <h3>Bus Information</h3>
        <div id="adminBusList"></div>

        <h3>Booking Report</h3>
        <button class="main-btn" onclick="showReport()">
          Generate Report
        </button>
        <div id="report"></div>
      </div>
    </section>

</div>

<footer>
  <p>Online Bus Ticket Booking System | Internet Programming Lab</p>
</footer>

<script>
  let users = [];
  let currentUser = null;
  let bookings = [];
  let comments = [];

  let buses = [
```

```
  {
    id: 1,
    name: "GreenLine Express",
    from: "Chennai",
    to: "Bengaluru",
    category: "Volvo AC",
    price: 800,
    seats: 20
  },
  {
    id: 2,
    name: "Night Rider",
    from: "Chennai",
    to: "Coimbatore",
    category: "Sleeper Coach",
    price: 650,
    seats: 20
  },
  {
    id: 3,
    name: "City Travels",
    from: "Bengaluru",
    to: "Chennai",
    category: "AC Coach",
    price: 750,
    seats: 20
  }
];

let selectedBus = null;
let selectedSeats = [];
```

```
document.getElementById("date").value =
  new Date().toISOString().split("T")[0];

document.getElementById("date").min =
  new Date().toISOString().split("T")[0];

function showPage(page) {
  document.querySelectorAll(".container section")
    .forEach(section => section.classList.add("hidden"));

  document.getElementById(page).classList.remove("hidden");

  if (page === "bookings") showBookings();
  if (page === "admin") showAdminBuses();
}

function registerUser() {
  let name = document.getElementById("name").value.trim();
  let email = document.getElementById("email").value.trim().toLowerCase();
  let password = document.getElementById("password").value;

  if (!name || !email || !password) {
    alert("Please fill in all fields");
    return;
  }

  if (users.some(user => user.email === email)) {
    alert("Email is already registered");
    return;
  }
```

```
    users.push({ name, email, password });
    currentUser = { name, email };

    document.getElementById("message").innerText =
      "Registration successful! You are logged in.";
}

function loginUser() {
    let email = document.getElementById("loginEmail").value.trim().toLowerCase();
    let password = document.getElementById("loginPassword").value;

    let user = users.find(
      user => user.email === email && user.password === password
    );

    if (user) {
      currentUser = { name: user.name, email: user.email };
      document.getElementById("message").innerText =
        "Login successful!";
    } else {
      alert("Invalid details. Please register first.");
    }
}

function searchBuses() {
    let from = document.getElementById("from").value;
    let to = document.getElementById("to").value;
    let date = document.getElementById("date").value;

    if (from === to) {
```

```
    alert("Choose different source and destination");
    return;
}

if (!date) {
    alert("Select your journey date");
    return;
}

let result = buses.filter(
    bus => bus.from.toLowerCase() === from.toLowerCase()
        && bus.to.toLowerCase() === to.toLowerCase()
);

let output = document.getElementById("busList");
output.innerHTML = "";

if (result.length === 0) {
    output.innerHTML = "<p>No buses found for this route.</p>";
    return;
}

result.forEach(bus => {
    let bookedSeats = bookings
        .filter(b => b.busId === bus.id &&
            b.date === date && b.status === "Confirmed")
        .flatMap(b => b.seats);

    output.innerHTML += `
        <div class="bus">
            <h3>${bus.name}</h3>
```

```
            <p>${bus.from} → ${bus.to}</p>
            <p>Category: ${bus.category}</p>
            <p>Price: ₹${bus.price}</p>
            <p>Available seats: ${bus.seats - bookedSeats.length}</p>
            <button class="main-btn"
              onclick="bookTicket(${bus.id})">
              Select Seats
            </button>
          </div>
      `;
    });
}

function bookTicket(busId) {
    if (!currentUser) {
      alert("Please register or log in first");
      showPage("register");
      return;
    }

    selectedBus = buses.find(bus => bus.id === busId);
    selectedSeats = [];

    let date = document.getElementById("date").value;

    let bookedSeats = bookings
      .filter(b => b.busId === busId &&
        b.date === date && b.status === "Confirmed")
      .flatMap(b => b.seats);

    let seatHTML = `<h3>Select seats for ${selectedBus.name}</h3>`;
```

```
  for (let i = 1; i <= selectedBus.seats; i++) {
    let isBooked = bookedSeats.includes(i);

    seatHTML += `
      <button class="seat ${isBooked ? "booked" : ""}"
        ${isBooked ? "disabled" : ""}
        onclick="selectSeat(${i}, this)">
        ${i}
      </button>
    `;
  }

  seatHTML += `
    <p id="selectedText">Selected seats: None</p>
    <p id="amountText">Total: ₹0</p>
    <input id="passengerName" placeholder="Passenger name">
    <input id="passengerAge" type="number" placeholder="Passenger age">
    <input id="passengerPhone" placeholder="Phone number">
    <input id="passengerEmail" type="email" placeholder="Confirmation email">
    <button class="main-btn" onclick="confirmBooking()">
      Pay & Confirm (Demo)
    </button>
  `;

  document.getElementById("busList").innerHTML =
    `<div class="bus">${seatHTML}</div>`;
}

function selectSeat(number, button) {
  if (selectedSeats.includes(number)) {
```

```
      selectedSeats = selectedSeats.filter(seat => seat !== number);
      button.classList.remove("selected");
    } else {
      selectedSeats.push(number);
      button.classList.add("selected");
    }

    document.getElementById("selectedText").innerText =
      "Selected seats: " + (selectedSeats.join(", ") || "None");

    document.getElementById("amountText").innerText =
      "Total: ₹" + (selectedSeats.length * selectedBus.price);
}

function confirmBooking() {
    if (!selectedSeats.length) {
      alert("Please select at least one seat");
      return;
    }

    let name = document.getElementById("passengerName").value.trim();
    let age = document.getElementById("passengerAge").value;
    let phone = document.getElementById("passengerPhone").value.trim();
    let email = document.getElementById("passengerEmail").value.trim();

    if (!name || !age || !phone || !email) {
      alert("Enter all passenger details");
      return;
    }

    let date = document.getElementById("date").value;
```

```
  let bookedSeats = bookings
    .filter(b => b.busId === selectedBus.id &&
      b.date === date && b.status === "Confirmed")
    .flatMap(b => b.seats);

  if (selectedSeats.some(seat => bookedSeats.includes(seat))) {
    alert("One or more seats are already booked. Search again.");
    searchBuses();
    return;
  }

  bookings.push({
    id: Date.now(),
    userEmail: currentUser.email,
    busId: selectedBus.id,
    busName: selectedBus.name,
    from: selectedBus.from,
    to: selectedBus.to,
    date: date,
    seats: [...selectedSeats],
    passenger: name,
    email: email,
    amount: selectedSeats.length * selectedBus.price,
    status: "Confirmed"
  });

  alert("Booking confirmed! Confirmation email is simulated.");
  searchBuses();
}
```

```
function showBookings() {
  let list = document.getElementById("bookingList");

  if (!currentUser) {
    list.innerHTML = "Please log in to view your bookings.";
    return;
  }

  let mine = bookings.filter(
    booking => booking.userEmail === currentUser.email
  );

  list.innerHTML = mine.length ? "" : "No bookings found.";

  mine.forEach(booking => {
    list.innerHTML += `
      <div class="bus">
        <h3>Booking ID: ${booking.id}</h3>
        <p>${booking.busName}: ${booking.from} → ${booking.to}</p>
        <p>Date: ${booking.date}</p>
        <p>Seats: ${booking.seats.join(", ")}</p>
        <p>Amount: ₹${booking.amount}</p>
        <p>Status: ${booking.status}</p>
        ${booking.status === "Confirmed" ?
          `<button class="main-btn"
            onclick="cancelBooking(${booking.id})">
            Cancel Booking
          </button>` : ""}
      </div>
    `;
  });
```

```
}

function cancelBooking(id) {
  let booking = bookings.find(b => b.id === id);

  if (booking && confirm("Cancel this ticket?")) {
    booking.status = "Cancelled";
    showBookings();
  }
}

function addComment() {
  let name = document.getElementById("commentName").value.trim();
  let text = document.getElementById("commentText").value.trim();

  if (!name || !text) {
    alert("Enter your name and comment");
    return;
  }

  comments.push({ name, text });
  document.getElementById("commentName").value = "";
  document.getElementById("commentText").value = "";
  showComments();
}

function showComments() {
  document.getElementById("commentList").innerHTML =
    comments.map(comment =>
      `<div class="bus"><b>${comment.name}</b><p>${comment.text}</p></div>`
    ).join("");
```

```
}

function adminLogin() {
  let user = document.getElementById("adminUser").value;
  let password = document.getElementById("adminPassword").value;

  if (user === "admin" && password === "admin123") {
    document.getElementById("adminPanel").classList.remove("hidden");
    showAdminBuses();
  } else {
    alert("Invalid admin username or password");
  }
}

function addBus() {
  let name = document.getElementById("busName").value.trim();
  let from = document.getElementById("busFrom").value.trim();
  let to = document.getElementById("busTo").value.trim();
  let category = document.getElementById("category").value;
  let price = Number(document.getElementById("price").value);
  let seats = Number(document.getElementById("totalSeats").value);

  if (!name || !from || !to || from.toLowerCase() === to.toLowerCase()
    || price <= 0 || seats <= 0) {
    alert("Enter valid bus information");
    return;
  }

  buses.push({
    id: Date.now(),
    name, from, to, category, price, seats
```

```
  });

  showAdminBuses();
  alert("Bus added successfully");
}

function showAdminBuses() {
  document.getElementById("adminBusList").innerHTML =
    buses.map(bus => `
      <div class="bus">
        <b>${bus.name}</b> — ${bus.from} to ${bus.to}
        <p>${bus.category} | ₹${bus.price} | ${bus.seats} seats</p>
        <button class="main-btn"
          onclick="deleteBus(${bus.id})">
          Delete Bus
        </button>
      </div>
    `).join("");
}

function deleteBus(id) {
  if (confirm("Delete this bus?")) {
    buses = buses.filter(bus => bus.id !== id);
    showAdminBuses();
  }
}

function showReport() {
  let confirmed = bookings.filter(b => b.status === "Confirmed");
  let revenue = confirmed.reduce((total, b) => total + b.amount, 0);
```

```
    document.getElementById("report").innerHTML = `
      <p>Total bookings: ${bookings.length}</p>
      <p>Confirmed bookings: ${confirmed.length}</p>
      <p>Cancelled bookings: ${bookings.length - confirmed.length}</p>
      <p>Total revenue: ₹${revenue}</p>
    `;
  }

  searchBuses();
</script>

</body>
</html>
```